

HOTA Metric — Compact Algorithm + Variable Reference

Compact Paper-Style Algorithm

Per Video / Sequence

Input

$$\{\mathcal{GT}_t, \mathcal{Pred}_t, S^t\}_{t=1}^T, \quad \mathcal{A} = \{0.05, 0.10, \dots, 0.95\}$$

Total G groundtruth tracks and P prediction tracks

1. Similarity normalization

For frame t , ground-truth detection i , and prediction j :

$$R_{ij}^t = \frac{S_{ij}^t}{\sum_{i=1}^{N_t} S_{ij}^t + \sum_{j=1}^{M_t} S_{ij}^t - S_{ij}^t}$$

Shape of R^t : $N_t \times M_t$

2. Accumulate global identity co-occurrence

For every GT identity g and predicted identity p :

$$C_{gp} = \sum_{t,i,j: ID_G(i)=g, ID_P(j)=p} R_{ij}^t.$$

Also count detections belonging to each identity over the whole video:

$$N_g^G = \sum_t \#\{i : ID_G(i) = g\},$$
$$N_p^P = \sum_t \#\{j : ID_P(j) = p\}.$$

Shape of C : $G \times P$

3. Global identity alignment

$$A_{gp} = \frac{C_{gp}}{N_g^G + N_p^P - C_{gp}}$$

Shape of A : $G \times P$

This score measures how strongly GT identity g aligns with predicted identity p over the complete video.

4. Frame-level matching score

For every pair (i, j) in frame t :

Per-frame quantities

Variable	Meaning	Shape
$\mathcal{GT}_t$	GT detections in frame t	set/list of size G_t
$\mathcal{Pred}_t$	Predicted detections in frame t	set/list of size P_t
S^t	Localization similarity matrix	$N_t \times M_t$
T	Number of frames in the video	scalar
N_t	Number of GT detections in frame t	scalar
M_t	Number of predicted detections in frame t	scalar
S_{ij}^t	Similarity between GT detection i and prediction j	scalar
$ID_G(i)$	GT track identity of detection i	scalar ID
$ID_P(j)$	Predicted track identity of detection j	scalar ID

Whole-video identity quantities

Variable	Meaning	Shape
G	Number of unique GT identities	scalar
P	Number of unique predicted identities	scalar
N^G	Total detection count for every GT identity	G
N_g^G	Number of GT detections belonging to identity g across the video	scalar
N^P	Total detection count for every predicted identity	P
N_p^P	Number of predictions belonging to identity p across the video	scalar
A	Global identity alignment matrix	$G \times P$

$$W_{ij}^t = A_{ID_G(i), ID_P(j)} \cdot S_{ij}^t$$

5. Optimal one-to-one matching

Apply Hungarian matching:

$$\Pi_t = \arg \max_{\Pi} \sum_{(i,j) \in \Pi} W_{ij}^t$$

subject to every GT detection and prediction being used at most once.

6. Threshold accepted matches

For every threshold $\alpha \in \mathcal{A}$:

$$\Pi_t^\alpha = \{(i,j) \in \Pi_t : S_{ij}^t \geq \alpha\}$$

Only Hungarian matches satisfying the original localization threshold are retained.

7. Detection counts

$$TP_\alpha = \sum_t |\Pi_t^\alpha|$$

$$FN_\alpha = \sum_t (N_t - |\Pi_t^\alpha|)$$

$$FP_\alpha = \sum_t (M_t - |\Pi_t^\alpha|)$$

8. Identity-pair match counts

For every GT identity g and predicted identity p :

$$M_{gp}^\alpha = \sum_{t=1}^T \# \{(i,j) \in \Pi_t^\alpha : ID_G(i) = g, ID_P(j) = p\}$$

Equivalently,

M_{gp}^α = number of accepted detection matches across the video
whose GT detection belongs to identity g and whose prediction belongs to identity p .
Thus,

$$M_{gp}^\alpha = 3$$

means GT track g and predicted track p were accepted as a matched pair in **three frames** at threshold α .

Variable	Meaning	Shape
A_{gp}	Alignment score between GT identity g and predicted identity p	scalar

Matching quantities

Variable	Meaning	Shape
W^t	HOTA matching-score matrix	$N_t \times M_t$
W_{ij}^t	Matching score $A_{gp} S_{ij}^t$	scalar
Π_t	Hungarian one-to-one matches in frame t	set of pairs
Π_t^α	Hungarian matches accepted after threshold α	set of pairs
α	Localization threshold	scalar
$\mathcal{A}$	Set of evaluation thresholds	19 scalars

Identity-pair match counts

Variable	Meaning	Shape
M^α	Accepted identity-pair match-count matrix	$G \times P$
M_{gp}^α	Number of accepted detection matches between GT identity g and prediction identity p across all frames	scalar

Intuition

If $M_{gp}^\alpha = 5$,
then five accepted detection pairs across the video connected GT identity g with predicted identity p at threshold α .
This quantity is therefore the bridge between **frame-level detection matching** and **track-level association quality**.

Per-threshold detection counts

Variable	Meaning	Shape
TP_α	Total accepted matched detections	scalar
FN_α	Total unmatched GT detections	scalar

9. Association Accuracy

For an identity pair (g, p) :

$$AssIoU_{gp}^\alpha = \frac{M_{gp}^\alpha}{N_g^G + N_p^P - M_{gp}^\alpha}$$

Then

$$AssA_\alpha = \frac{\sum_{g,p} M_{gp}^\alpha AssIoU_{gp}^\alpha}{\max(1, TP_\alpha)}$$

or directly,

$$AssA_\alpha = \frac{\sum_{g,p} M_{gp}^\alpha \frac{M_{gp}^\alpha}{N_g^G + N_p^P - M_{gp}^\alpha}}{\max(1, TP_\alpha)}.$$

10. Localization Accuracy

Accumulate localization similarity of accepted matches:

$$L_\alpha = \sum_t \sum_{(i,j) \in \Pi_t^\alpha} S_{ij}^t.$$

Then

$$LocA_\alpha = \frac{\max(10^{-10}, L_\alpha)}{\max(10^{-10}, TP_\alpha)}$$

If TP=0, then LocA = 0

11. Association Recall and Precision

$$AssRe_\alpha = \frac{\sum_{g,p} M_{gp}^\alpha \frac{M_{gp}^\alpha}{N_g^G}}{\max(1, TP_\alpha)}$$

$$AssPr_\alpha = \frac{\sum_{g,p} M_{gp}^\alpha \frac{M_{gp}^\alpha}{N_p^P}}{\max(1, TP_\alpha)}$$

12. Detection metrics

$$DetRe_\alpha = \frac{TP_\alpha}{TP_\alpha + FN_\alpha}$$

$$DetPr_\alpha = \frac{TP_\alpha}{TP_\alpha + FP_\alpha}$$

$$DetA_\alpha = \frac{TP_\alpha}{TP_\alpha + FN_\alpha + FP_\alpha}$$

Variable	Meaning	Shape
FP_α	Total unmatched predicted detections	scalar
L_α	Sum of localization similarities of accepted matches	scalar

Per-threshold metrics

Variable	Meaning	Shape
$AssIoU_{gp}^\alpha$	Association IoU/Jaccard for identity pair (g, p)	scalar
$AssA_\alpha$	Association Accuracy	scalar
$AssRe_\alpha$	Association Recall	scalar
$AssPr_\alpha$	Association Precision	scalar
$LocA_\alpha$	Localization Accuracy	scalar
$DetRe_\alpha$	Detection Recall	scalar
$DetPr_\alpha$	Detection Precision	scalar
$DetA_\alpha$	Detection Accuracy	scalar
$HOTA_\alpha$	HOTA at threshold α	scalar

Multi-video quantities

Variable	Meaning	Shape
K	Number of videos/sequences	scalar
k	Video index	scalar
$TP_\alpha^{(k)}$	TP count from video k	scalar
$FN_\alpha^{(k)}$	FN count from video k	scalar
$FP_\alpha^{(k)}$	FP count from video k	scalar
$AssA_\alpha^{(k)}$	Association Accuracy of video k	scalar
$AssRe_\alpha^{(k)}$	Association Recall of video k	scalar
$AssPr_\alpha^{(k)}$	Association Precision of video k	scalar
$LocA_\alpha^{(k)}$	Localization Accuracy of video k	scalar

13. HOTA

$$HOTA_{\alpha} = \sqrt{DetA_{\alpha} \cdot AssA_{\alpha}}$$

Combine Multiple Videos / Sequences

Let the dataset contain

$$V_1, V_2, \dots, V_K.$$

1. Sum detection counts

$$TP_{\alpha} = \sum_{k=1}^K TP_{\alpha}^{(k)}$$

$$FN_{\alpha} = \sum_{k=1}^K FN_{\alpha}^{(k)}$$

$$FP_{\alpha} = \sum_{k=1}^K FP_{\alpha}^{(k)}.$$

2. TP-weighted association metrics

For

$$X \in \{AssA, AssRe, AssPr\},$$

$$X_{\alpha} = \frac{\sum_{k=1}^K TP_{\alpha}^{(k)} X_{\alpha}^{(k)}}{TP_{\alpha}}$$

3. TP-weighted localization accuracy

$$LocA_{\alpha} = \frac{\sum_{k=1}^K TP_{\alpha}^{(k)} LocA_{\alpha}^{(k)}}{TP_{\alpha}}$$

4. Recompute dataset detection metrics

$$DetRe_{\alpha} = \frac{TP_{\alpha}}{TP_{\alpha} + FN_{\alpha}}$$

$$DetPr_{\alpha} = \frac{TP_{\alpha}}{TP_{\alpha} + FP_{\alpha}}$$

$$DetA_{\alpha} = \frac{TP_{\alpha}}{TP_{\alpha} + FN_{\alpha} + FP_{\alpha}}$$

5. Recompute dataset HOTA

$$HOTA_{\alpha} = \sqrt{DetA_{\alpha} \cdot AssA_{\alpha}}$$

Also:

$$OWTA_{\alpha} = \sqrt{DetRe_{\alpha} \cdot AssA_{\alpha}}.$$

OWTA score focuses only on matched detections by not including FP's

Final reported HOTA

$$HOTA = \frac{1}{19} \sum_{\alpha=0.05}^{0.95} HOTA_{\alpha}$$

where α increases in steps of 0.05.